

Exercise (1)

N. goals	freq	cf	M_i	$(M_i - \bar{x})$	$(M_i - \bar{x})^2$	$f_i (M_i - \bar{x})^2$
10-12	4	4	11	5,65	31,92	127,6
13-15	12	16	14	-2,65	7,02	84,27
16-18	22	38	17	0,35	0,12	2,69
19-21	14	52	20	3,35	11,22	157,11
total	52					371,77

1) the characteristics of two sets of data

* The Q_i

$$Q_i = L_{q_i} + \left(\frac{\frac{i \times n}{4} - c_{bq_i}}{f_{q_i}} \right) \times w$$

* for $(i=1)$: $\frac{i \times n}{4} = \frac{1 \times 52}{4} = 13 \in [13, 15]$

$L_{q_i} = 13$; $c_{bq_i} = 4$; $f_{q_i} = 12$; $n = 52$; $w = 2$.

$\Rightarrow Q_1 = 13 + \left(\frac{\frac{52}{4} - 4}{12} \right) \times 2 = 14,50$

* for $(i=2)$: $\frac{i \times n}{4} = \frac{2 \times 52}{4} = 26 \in [16, 18]$

$L_{q_i} = 16$; $c_{f_i} = 16$; $f_{q_i} = 22$; $n = 52$; $w = 2$.

$\Rightarrow Q_2 = 16 + \left(\frac{\frac{2 \times 52}{4} - 16}{22} \right) \times 2 = 16,9 = \text{median}$

* for $(i=3)$: $\frac{i \times n}{4} = \frac{3 \times 52}{4} = 39 \in [19, 21]$

$L_{q_i} = 19$; $c_{f_i} = 38$; $f_{q_i} = 14$; $n = 52$; $w = 2$.

$\Rightarrow Q_3 = 19 + \left(\frac{\frac{3 \times 52}{4} - 38}{14} \right) \times 2 = 19,14$

* the means

$\bar{X} = \frac{\sum_{i=1}^n f_i \cdot M_i}{n} = \frac{11 \times 4 + 14 \times 12 + 17 \times 22 + 20 \times 14}{52} = 16,65$

* the Var: $s^2 = \frac{1}{n \rightarrow \sum_{i=1}^n f_i - 1} \sum_{i=1}^n f_i (x_i - \bar{x})^2 = \frac{1}{51} (371,77) \Rightarrow \boxed{s^2 = 7,27}$

* the STD: $STD = s = \sqrt{s^2} = \sqrt{7,27} = \boxed{2,69}$

2) The absolute deviation

$$AD = \frac{1}{\sum_{i=1}^n f_i} \sum_{i=1}^n f_i |M_i - \bar{x}| = \frac{4 \times 5,65 + 12 \times 2,65 + 22 \times 0,35 + 14 \times 3,35}{52} \\ \Rightarrow AD = \boxed{2,09}$$

3) the new mean

$$\bar{x} = \frac{\sum_{i=1}^n f_i \times m_i (m_i' = 2m_i)}{n} = \frac{\sum_{i=1}^n f_i \times (2m_i)}{n} = 2 \frac{\sum_{i=1}^n f_i \times m_i}{n} = \boxed{2\bar{x}}$$

* the new STD

$$Var' = \frac{1}{\sum_{i=1}^n f_i} \sum_{i=1}^n f_i (m_i' - \bar{x}')^2 \quad \begin{matrix} \nearrow \bar{x}' = 2\bar{x} \\ \searrow m_i' = 2m_i \end{matrix} \Rightarrow Var' = \frac{1}{\sum_{i=1}^n f_i} \sum_{i=1}^n f_i (2m_i - 2\bar{x})^2 \\ = 2^2 \left(\frac{1}{\sum_{i=1}^n f_i} \sum_{i=1}^n f_i (m_i - \bar{x})^2 \right) \Rightarrow Var' = \boxed{4Var}$$

$$\Rightarrow STD' = \sqrt{Var'} = \sqrt{4Var} = 2\sqrt{Var} \Rightarrow STD' = \boxed{2STD}$$

Exercise (2)

1) the percentage of scores lie within one standard deviation from the mean

$$I = [\bar{x} - K \times SD ; \bar{x} + K \times SD] \quad | \quad 1 - \frac{1}{K^2}$$

$$\bar{x} = \frac{\sum_{i=1}^{20} x_i \times f_i}{N} = \boxed{84,5}$$

$$Var = \frac{1}{\sum_{i=1}^n f_i - 1} \times \sum_{i=1}^n f_i (x_i - \bar{x})^2 = \boxed{164,75}$$

$$SD = \sqrt{164,75} = \boxed{12,83}$$

* for (K=1): $I = [84,5 - 1 \times 12,83 ; 84,5 + 1 \times 12,83] = [71,67 ; 97,33]$

$$1 - \frac{1}{K^2} = 1 - \frac{1}{1^2} = \boxed{0\%}$$

* for (K=2): $I = [84,5 - 2 \times 12,83 ; 84,5 + 2 \times 12,83] = [58,82 ; 110,16]$

$$1 - \frac{1}{K^2} = 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = 0,75 = \boxed{75\%}$$

p) $\frac{19}{20} \times 100 = 95\% \Rightarrow$ so Chebyshev's theorem is satisfied.

Exercise (3)

1) the box plots.

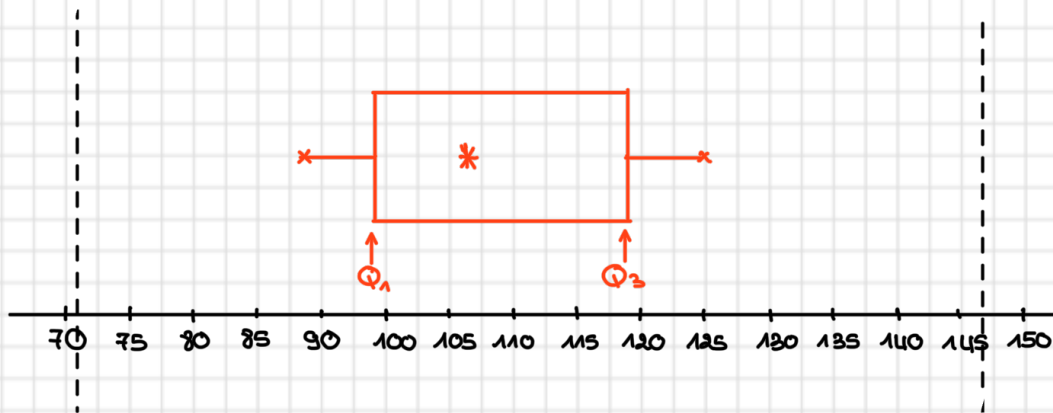
- sample 1: 87-99-100-105-105-107-108-118-119-125.

$$Q_1^{(p)} = \left(\frac{n+1}{4}\right)^{th} = \frac{11}{4} = 2,75 \notin \mathbb{N} \text{ then } Q_1 = \frac{X_{(2)} + X_{(3)}}{2} = \frac{99 + 100}{2} = 99,5$$

$$Q_2^{(p)} = 2 \times \left(\frac{n+1}{4}\right)^{th} = 5,5 \notin \mathbb{N} \text{ then } Q_2 = \frac{X_{(5)} + X_{(6)}}{2} = \frac{105 + 107}{2} = 106 \text{ [median]}$$

$$Q_3^{(p)} = 3 \times \left(\frac{n+1}{4}\right)^{th} = 8,25 \notin \mathbb{N} \text{ then } Q_3 = \frac{X_{(8)} + X_{(9)}}{2} = \frac{118 + 119}{2} = 118,5$$

$$\begin{aligned} \text{IQR} &= Q_3 - Q_1 = 118,5 - 99,5 = 19 \\ Q_1 - \text{IQR} \times 1,5 &= 71 \\ Q_3 + \text{IQR} \times 1,5 &= 147 \end{aligned}$$



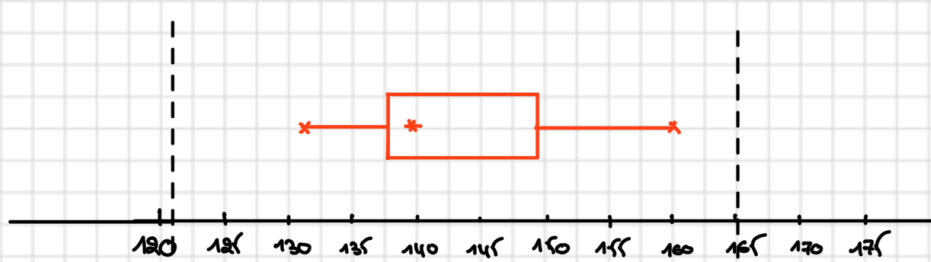
- sample 2: 133-137-138-138-139-140-142-145-152-160.

$$Q_1^{(p)} = \left(\frac{n+1}{4}\right)^{th} = \frac{11}{4} = 2,75 \notin \mathbb{N} \text{ then } Q_1 = \frac{X_{(2)} + X_{(3)}}{2} = \frac{137 + 138}{2} = 137,5$$

$$Q_2^{(p)} = 2 \times \left(\frac{n+1}{4}\right)^{th} = 5,5 \notin \mathbb{N} \text{ then } Q_2 = \frac{X_{(5)} + X_{(6)}}{2} = \frac{139 + 140}{2} = 139,5$$

$$Q_3^{(p)} = \frac{X_{(8)} + X_{(9)}}{2} = \frac{145 + 152}{2} = 148,5$$

$$\begin{aligned} \text{IQR} &= Q_3 - Q_1 = 148,5 - 137,5 = 11 \\ Q_1 - \text{IQR} \times 1,5 &= 121 \\ Q_3 + \text{IQR} \times 1,5 &= 165 \end{aligned}$$



2) Pearson's coefficient of skewness to confirm if these distributions are symmetric =

$$SK = \frac{3(\bar{X} - \tilde{X})}{SD} \quad SD = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

* Sample (1):

$$\bar{X}(\text{mean}) = \frac{87 + 99 + 100 + 105 \times 2 + 107 + 108 + 118 + 119 + 125}{10} = 107,3$$

$$\tilde{X}(\text{median}) = 106$$

$$SD = \sqrt{\frac{1}{9} \times [412,09 + 67,89 + 53,29 + 5,29 + 5,29 + 0,09 + 0,49 + 114,49 + 136,89 + 313,29]}$$

$$\Rightarrow SD = 11,10$$

$$So \text{ } SK = \frac{3 \times (107,3 - 106)}{11,10} = 0,35 \text{ . it is close to 0 which means it's almost symmetric.}$$

* Sample (2):

$$\bar{X}(\text{mean}) = \frac{133 + 137 + 138 \times 2 + 139 + 140 + 142 + 145 + 152 + 160}{10} = 142,4$$

$$\tilde{X}(\text{median}) = 139,5$$

$$So \text{ } SD = \sqrt{\frac{1}{9} \times [88,36 + 29,16 + 19,36 + 19,36 + 11,56 + 5,76 + 0,16 + 6,76 + 92,16 + 309,76]}$$

$$\Rightarrow SD = 8,0443$$

$$So \text{ } SK = \frac{3 \times (142,4 - 139,5)}{8,0443} = 1,081 \text{ . moderate positive skewness.}$$

* Sample (3):

$$\bar{X} = \frac{60 + 74 + 79 + 80 \times 2 + 82 \times 4 + 83}{10} = 78,4$$

$$\tilde{X} = 81$$

$$SD = \sqrt{\frac{1}{9} \times [174,96 + 761,76 + 1062,76 + 1128,96 \times 2 + 1267,36 \times 4 + 1339,56]}$$

$$\Rightarrow SD = 34,44$$

$$So \text{ } SK = \frac{3 \times (78,4 - 81)}{34,44} = -1,12 \text{ . considerable negative skewness}$$

* sample (1) is app symmetric.

3) the kind of fishes has the lowest variation in length.

$$CV = \frac{s}{\bar{x}} \times 100$$

* sample (1): $CV_1 = \frac{11,1}{107,3} \times 100 = 10,34\%$

* sample (2): $CV_2 = \frac{8,04}{139,5} \times 100 = 5,76\%$

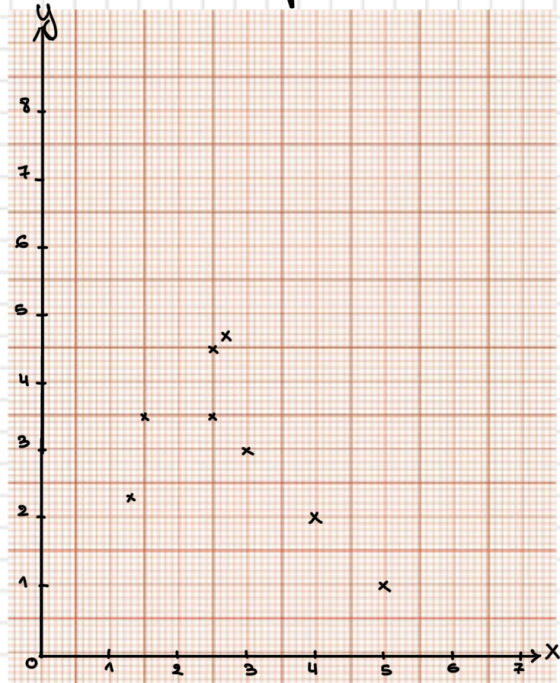
* sample (3): $CV_3 = \frac{6,96}{81} \times 100 = 8,59\%$

$$\Rightarrow CV_2 < CV_3 < CV_1$$

so sample (2) has the lowest variation length.

Exercise (4)

1) the scatterplot



2) the covariance and the linear correlation coefficient

$$cov(x; y) = \frac{1}{N} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$P(x; y) = \frac{cov(x; y)}{SD_x \cdot SD_y}$$

$$\bar{x} = \frac{\sum x_i}{N} = \frac{22,5}{8} = 2,81$$

$$\bar{y} = \frac{\sum y_i}{N} = \frac{24,5}{8} = 3,06$$

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
1,3	2,3	-1,51	-0,76	1,1476
1,5	3,5	-1,31	0,44	-0,5764
2,5	3,5	-0,31	0,44	-0,1364
2,5	4,5	-0,31	1,44	-0,4464
2,7	4,7	-0,11	1,64	-0,1804
3	3	0,19	-0,06	-0,0114
4	2	1,19	-1,06	-1,2614
5	1	2,19	-2,06	-4,5114
$\Sigma = 22,5$	$\Sigma = 24,5$	$\Sigma = 0,02$	$\Sigma = 0,02$	$\Sigma = -5,9762$

$$\ast \text{cov}(x, y) = \frac{1}{8} \times (-5,9762) \Rightarrow \boxed{\text{cov}(x, y) = -0,747025}$$

$$\ast SD_x = \sqrt{\frac{10,4487}{8}} \approx \boxed{1,1428} \quad \left| \quad \ast SD_y = \boxed{1,1779} \right.$$

$$\ast \rho(x, y) = \frac{-0,74}{1,14 \times 1,17} \Rightarrow \rho(x, y) = \boxed{-0,555}$$

$\Rightarrow$ since $\text{cov}(x, y) < 0$ then the relation between x and y is negative

$\Rightarrow$ since $\rho(x, y) < 0$ then the relation is moderate negative linear correlation relation.

Exercise (5):

1) marginal mean of (x): $[E(x)]$.

$$\bar{X}_{\text{marg}} = \frac{\sum n_i x_i}{\sum n_i} = \frac{2105 + 3217,5 + 3035}{533} = \frac{192}{533} \bar{X}_A + \frac{241}{533} \bar{X}_B + \frac{100}{533} \bar{X}_C = \boxed{15,68}$$

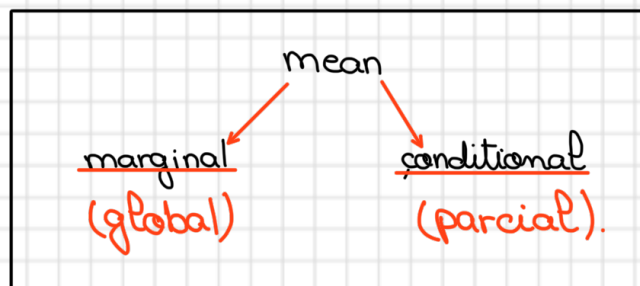
conditional mean of (x):

notation: $\bar{X}_A = E(x|y=A)$

$$\bar{X}_A = \frac{2105}{192} = \boxed{10,96}$$

$$\bar{X}_B = \frac{3217,5}{241} = \boxed{13,35}$$

$$\bar{X}_C = \frac{3035}{100} = \boxed{30,35}$$



2) the marginal variance of (x):

$$\text{Var}(x) = \frac{\sum n_i x_i^2}{N} - (\bar{x})^2 = \frac{31750 + 50706 + 94175}{533} - (15,68)^2 = \boxed{85,52}$$

conditional variance of (x):

$$\text{Var}_A(x) = \frac{31750}{192} - (10,96)^2 = \boxed{45,1662} \Rightarrow \text{Var}(x|y=A).$$

$$\text{Var}_B(x) = \frac{50706,25}{241} - (13,35)^2 = \boxed{32,1608} \Rightarrow \text{Var}(x|y=B).$$

$$\text{Var}_C(x) = \frac{94175}{100} - (30,35)^2 = \boxed{20,6275} \Rightarrow \text{Var}(x|y=C).$$

3) the correlation between the 3 modalities of y pairwise (two by two).

A,B B,C A,C

use the arithmetic mean!

$$\rho(A, B) = \frac{\text{cov}(A, B)}{SD_A \times SD_B} = \frac{49,19,666}{5790,79} \approx \boxed{0,8496}$$

↳ strong positive correlation.

Exercise (6)

$n = 164$

1) calculating the correlation coefficient

$$\rho(x, y) = \frac{\text{Cov}(x, y)}{SD_x \times SD_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$\bar{x} = \frac{\sum n_i x_i}{n} = 43,4$$

$$\bar{y} = \frac{\sum n_i y_i}{n} = 53,633$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 1799128$$

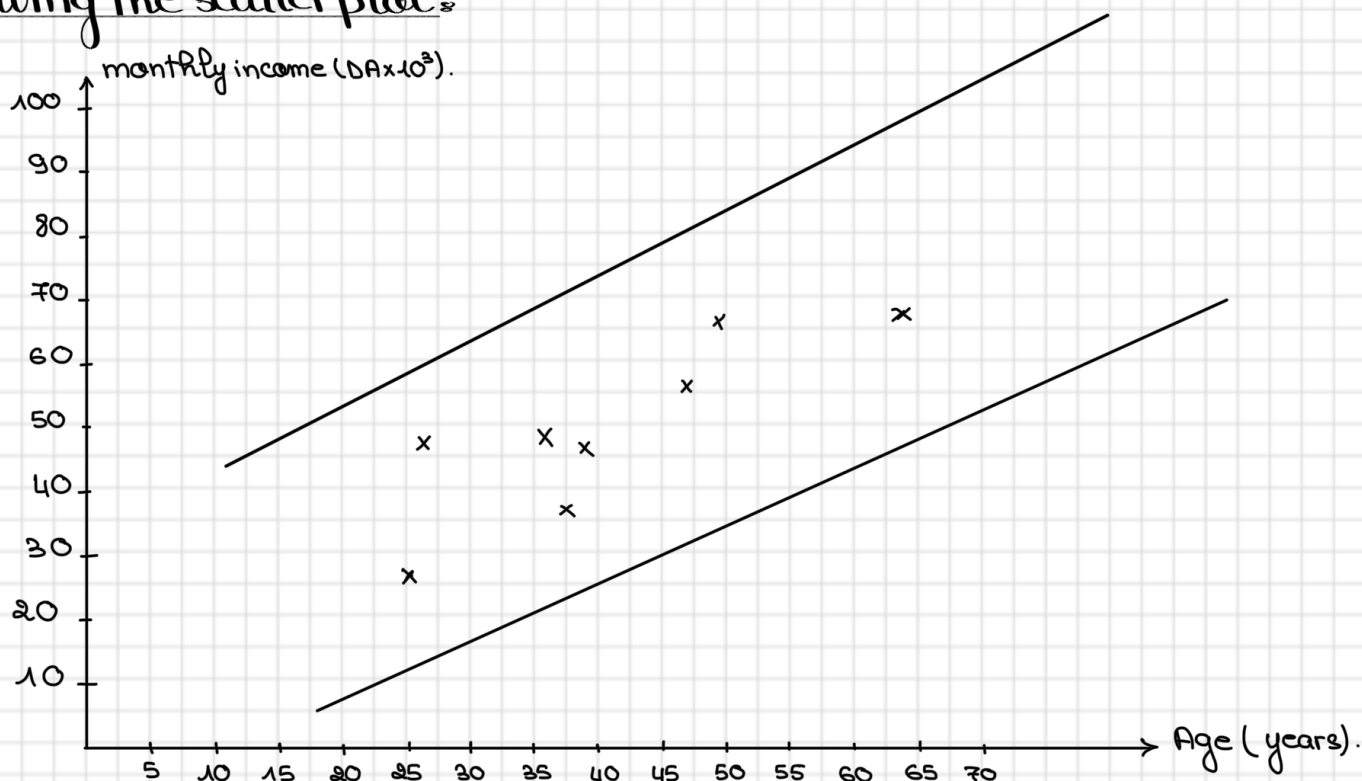
$$\rho(x, y) = \frac{1799128}{\sqrt{1274,4 \times 320117410}} \approx 0,85$$

$$\sum (x_i - \bar{x})^2 = 1274,4$$

$$\sum (y_i - \bar{y})^2 = 320117410$$

⇒ since $\rho(x, y) = 0,85$ so it's very strong positive correlation.

2) Drawing the scatter plot



3) the cross tabulation

Health. cond	R ₁	freq	R ₂	d _i = R ₁ - R ₂	d _i ²
Very good.	2,5	22	3	-0,5	0,25
Very good.	2,5	22	3	-0,5	0,25
good.	4,5	15	6	-1,5	2,25
fair.	6,5	16	5	1,5	2,25
Excellent.	1	20	1	0	0
Poor.	9	5	10	-1	1
Poor.	9	10	8,5	0,5	0,25
Poor.	9	10	8,5	0,5	0,25
Good.	4,5	22	3	1,5	2,25
fair.	6,5	13	7	-0,5	0,25

⇒ order them and do R₁ and R₂ according to it. (rank).

Σ = 9

$$r_s = 1 - \frac{6 \left[\sum d_i^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \dots \right]}{N(N^2-1)} = 1 - \frac{6 \left[9 + \frac{2(2^2-1)}{12} + \frac{2(2^2-1)}{12} + \frac{2(2^2-1)}{12} + \frac{2(2^2-1)}{12} + \frac{3(3^2-1)}{12} + \frac{3(3^2-1)}{12} \right]}{10(10^2-1)}.$$

$$= 1 - \frac{6[9+6]}{990} \Rightarrow \boxed{r_s = 0,9}$$